Introduction to Data Analytics
MS - 4610

Session 14 – Clustering

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Plan for the session

- 6 micro-cases to motivate clustering
- Overview of the different types of clustering/clusters
- Deep dive into two algorithms
  - K-means
  - Hierarchical
- Cluster evaluation through silhouettes and size determination through gap-statistic

Understanding clustering through micro-cases

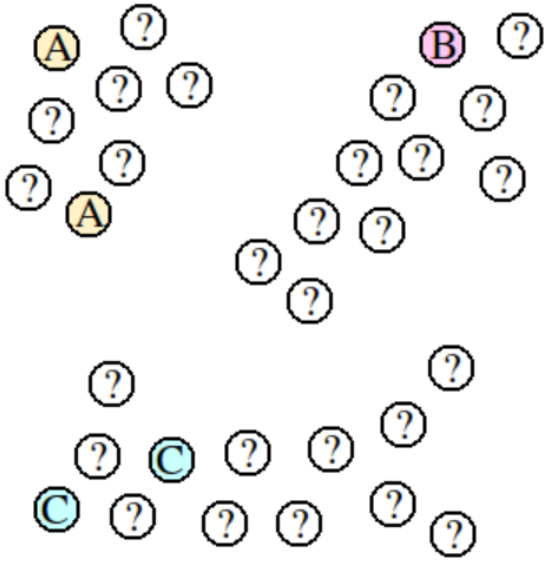
- The Bell computers case
  - Laptops are used for different purposes
  - Laptops are configurable in an online store (RAM, CPU, GPU, CD-Drive, ports, warranty)
  - Can we use historic purchase choices to infer the usage of the product?
- The AirCom case
  - One of the oldest telecom service providers
  - The migration to mobile phones, intro of 3G/4G brought in a lot of competition and niche entrants. Resulting in decline of market share.
  - The organization needs to understand what product plans are desirable (profitable is another question)
  - At a customer level, can we study the current call usage, sms, data? Frequency, quantum, night/day, weekend/weekday

Understanding clustering though micro-cases

- The Gmail case
  - By now we understand that:
    - Divides data into groups (clusters or segments or partitions)
    - Unsupervised learning
  - Why is it different from classification? The Gmail spam example
- More cases:
    - Marketing/Sales. Know your customer. Difference between target and consumption audience. Stories from Ford and the make-up industry.
    - Communicating information – Google News, sentiment analysis.
    - Biology, Climate, Medicine.

Introduction to Clustering

- Why do this?
  - For better understanding. For better processing by subject-matter experts.
  - Could serve as precursor to further Data Analysis. Labelled data is hard to get. Take the following classification example:



Wikipedia contributors. "Transduction (machine learning)." *Wikipedia, The Free Encyclopedia*. Wikipedia, The Free Encyclopedia, 5 May. 2016. Web. 19 Oct. 2018.

Types of Clustering and Clusters

- Types of Clustering
  - Exclusive versus Overlapping versus Fuzzy
  - Hierarchical versus Partitional
  - Complete versus Partial
- Types of Clusters
  - Well separated
  - Prototype Based
  - Graph based – types of networks
  - Density based

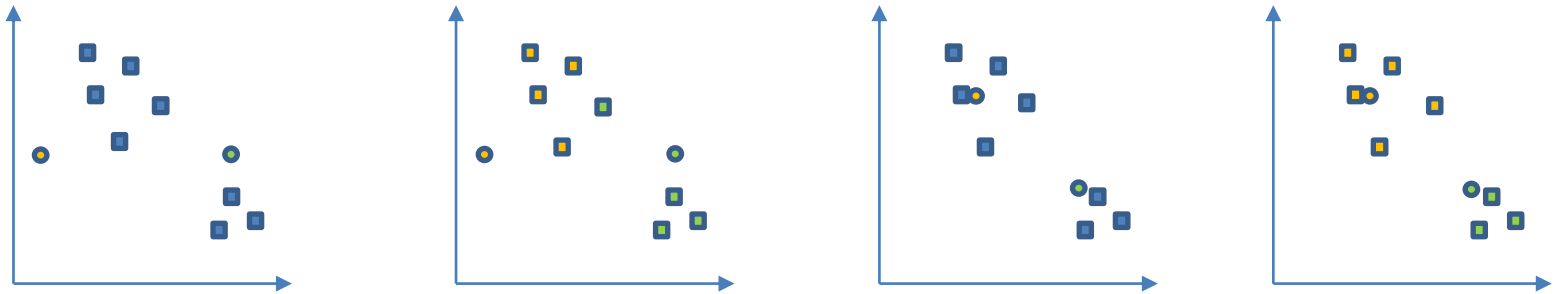
K-Means Clustering

- What is it?
  - Prototype based approach
  - An iterative procedure that starts with K clusters
  - Ideal when all variables are quantitative (Should be able to compute distances)
- How does it work?

1. Initialize some cluster centers (K)
2. Assign each point to the closest cluster centre
3. Once all points have been assigned, recompute the cluster centre to be the centroid of the assignment derived from step 2
4. Repeat steps 2 and 3 until no further changes in centroid

K-means and K-medoids

- K-means Graphically



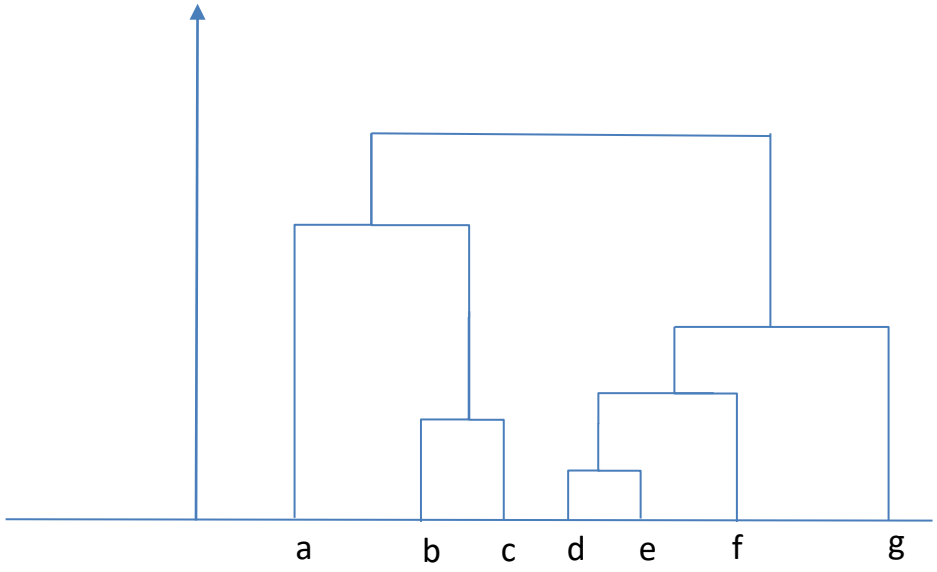
- Extending K-means to K-medoids
  - Going beyond quantitative variables to arbitrarily defined dissimilarities
  - Computational intensity

Hierarchical Clustering

- Does not require a fixed number of clusters
- Creates nested clusters
- Types:
  - Agglomerative (Bottom-up)
  - Divisive (Top-down)
- Requires defining some measure of dissimilarity between clusters
  - Single Linkage
  - Complete Linkage
  - Group Average
  - Ward's method, increase in SSE from a merger

Agglomerative Hierarchical Clustering

- Monotonicity
- Graphical representation through Dendrogram

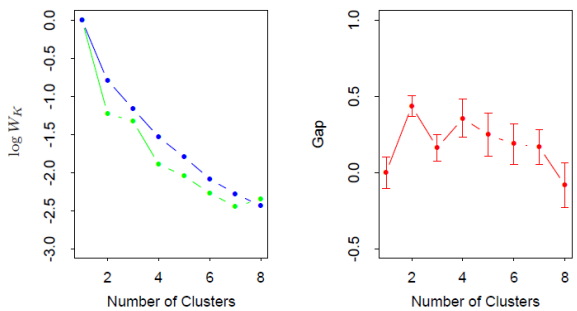


How many Clusters?

- Cluster Evaluation. Why do we need to evaluate?
- Internal versus External (unsupervised versus supervised). Would Cross validation work?
- Internal: cohesion versus separation
- Silhouettes:
  - For each data point i define $a(i)$ as the average dissimilarity of i with the other data points in its cluster. SSE should work.
  - For the same data point i define the vector $\mathrm{c}(i)$ as the average dissimilarity of i with the other data points in each cluster. Define $\mathrm{b(i)} = \min(\mathrm{c}(i))$
  - $s(i) = \frac{b(i)-a(i)}{\max(a(i),b(i))}$
  - Average s (across all datapoints) tells us something about the clustering. What is the range of s? what are desirable values?

Gap Statistic

- The problem with Silhouettes for a wide range of k (natural bias towards high k)
- Say you had some measure of within cluster dissimilarity W_K, which is a function of k (number of clusters), for a given repeatable clustering algorithm.
- As K goes from 1,2…to…N , the values of $W_1, W_2, \ldots, W_N$ tend to generally decrease.
- What if you assumed that the data was uniformly distributed?



Friedman, Jerome, Trevor Hastie, and Robert Tibshirani. The elements of statistical learning. Vol. 1. New York: Springer series in statistics, 2001.